

ORIGINAL ARTICLE

Modeling the influence of rhizodeposits on root water uptake

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Abstract

The chemical compounds that plant roots release into the soil, referred to as rhizodeposits, affect soil hydraulic properties like the surface tension and viscosity of soil water, and the contact angle between menisci and the pore surface. What remains less clear is how these effects manifest when considering soil water infiltration and retention, and the consequent impact on the availability of water for uptake by roots. A novel model for soil water transport was developed that incorporates the influences of rhizodeposits on soil hydraulics and root water uptake. The finite element method was used to generate model simulations and investigate how the presence of rhizodeposits of wheat (*Triticum aestivum* L.) affected water uptake of three differently aged root systems under precipitation regimes that varied in total precipitation and delivery pattern. The overriding effect on soil hydraulics of wheat rhizodeposits was to facilitate infiltration. A trade-off was identified in how this could improve water availability to the root system by reducing evaporation but also worsen it by increasing deep percolation. When total precipitation was shared over several rainfall events, the effect of rhizodeposits on root water uptake was positive. However, when higher levels of total precipitation were delivered in a single event, the presence of rhizodeposits reduced the water uptake of shallower root systems. This suggests that the influence of rhizodeposits on soil hydraulics and root water uptake is nuanced and will require careful consideration when developing crops for improved water use efficiency and stress resilience.

Plain Language Summary

The chemical compounds that plant roots release into the soil are referred to generally as rhizodeposits and these are known to affect the way water moves through the soil. This work aimed to investigate how rhizodeposit-induced changes to soil water movement impact the availability of water for uptake by plant roots. First, a mathematical model for soil water transport was developed that incorporates the influence of rhizodeposits. Simulations from this model were then used to assess the effect of rhizodeposits on the water uptake performance of different ages of wheat root systems under various rainfall conditions. Within the simulations, it was found that

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rhizodeposit presence could either improve or worsen the availability of water to plant roots, and that this depended on rainfall pattern and root system age. This suggests that rhizodeposit characteristics require careful consideration when developing crops for improved performance within specific environments.

1 | INTRODUCTION

Improving plant water use efficiency is a central aim of agricultural research and, as previously outlined by Condon et al. (2004), the water use efficiency of a plant is defined by three factors. The first is its ability to convert transpired water into biomass, the second is maximizing the proportion of this biomass that then constitutes harvestable product, and the final aspect (the focus of this paper) is the capacity of the plant's root system to uptake as much water as possible. Within the context of the final facet, much scientific attention has been dedicated to determining the root system architecture (RSA) that optimizes a plant's access to the regions of soil with the highest water content. Historically, the most promoted morphology is one where the majority of resources are allocated to growing downward so that stores of water deep in the soil can be accessed during periods of drought (Lynch, 2013; Lynch & Wojciechowski, 2015; Uga et al., 2013). Such root systems are less exposed to volatile rainfall patterns and, hence, more resilient in intensive agriculture systems (Wasson et al., 2012).

In addition to the provision of access to stored soil water, it is also important to consider the way in which the root system alters soil hydraulics, and how this affects the availability of water to the plant. Many studies have found that the presence of roots in soil facilitates water infiltration following precipitation (Cerdeira, 1999; Marshall et al., 2014; Wu et al., 2016). Indeed, for a range of soil types, in comparison to being left fallow, their hydraulic conductivities have been found to be higher when vegetated by crops such as maize, willow, and grass (Feki et al., 2018; Leung et al., 2018). The proposed mechanisms behind this are related to the preferential flow of soil water that occurs as the structure and hydraulic properties of the rhizosphere, the soil surrounding plant roots, are altered due to the penetration of growing roots and the activity of the fauna that their by-products attract (Angers & Caron, 1998; Luo et al., 2019). Therefore, the RSA of a plant not only determines where water can be extracted from, but also influences the speed and direction of water movement through the soil following irrigation or rainfall (Ghestem et al., 2011; Noguchi et al., 1997, 1999). From this perspective, a desirable root system would be one that promotes a pattern of infiltration whereby the provision of water to the rooted zone is facilitated and losses of available water through evaporation and excessive drainage are minimized (Mair et al., 2023). As illustrated

in Figure 1a, however, it is not the RSA alone that determines this influence on soil water dynamics. Depending on species, between 25% and 70% of the fixed carbon that is allocated by crops to their root systems is released into the soil in the form of rhizodeposits (McGrail et al., 2020). Across a range of crop types, these chemical compounds are reported to affect a number of soil hydraulic properties. Specific observed effects include a decrease in the surface tension of the soil water solution as the concentration of wheat, maize, or barley rhizodeposit is increased (Naveed et al., 2019; Read et al., 2003; Read & Gregory, 1997), and an increase in its viscosity with higher levels of maize, barley, or wheat rhizodeposit concentration (Hosseini and Mosaddeghi, 2024; Naveed et al., 2019). Additionally, higher concentrations of maize and wheat rhizodeposits adsorbed to the soil particle surfaces have been shown to increase the contact angle between pore water and the pore surface (Ahmed et al., 2016; Benard et al., 2018; Naveed et al., 2019; Zickenrott et al., 2016). It is reasonable then to conclude that these different rhizodeposit effects must also play a role in determining how water transport through a vegetated soil is influenced by plant roots. However, due to the challenges in observing rhizosphere processes under controlled field conditions (Oburger & Jones, 2018), it has proved difficult to formally establish this relationship experimentally and reveal any likely consequences regarding the availability of water for uptake by a root system (Preece & Peñuelas, 2016).

Therefore, the use of mathematical models of soil hydraulics is critical for advancing our understanding of how plants influence soil water dynamics, and how this affects their capacity to access water in the soil. The standard starting point when developing a model for water transport through soil is Richards equation (L. A. Richards, 1931), a partial differential equation (PDE) where the flux of soil water is driven by gravity and the negative gradient in pressure. Within this model are functions for the nonlinear relations between pressure, soil water content, and hydraulic conductivity (Brooks & Corey, 1964; Mualem, 1976; van Genuchten, 1980), and it is through alterations to the parametrization of these functions that the effect of rhizodeposits on soil hydraulics tends to be incorporated (Carminati et al., 2016). It has been shown by Karagunduz et al. (2001, 2015) that a scaling of the pressure head term in the van Genuchten (1980) water retention function allowed their model to accurately reproduce experimental observations on the effect on infiltration of changes

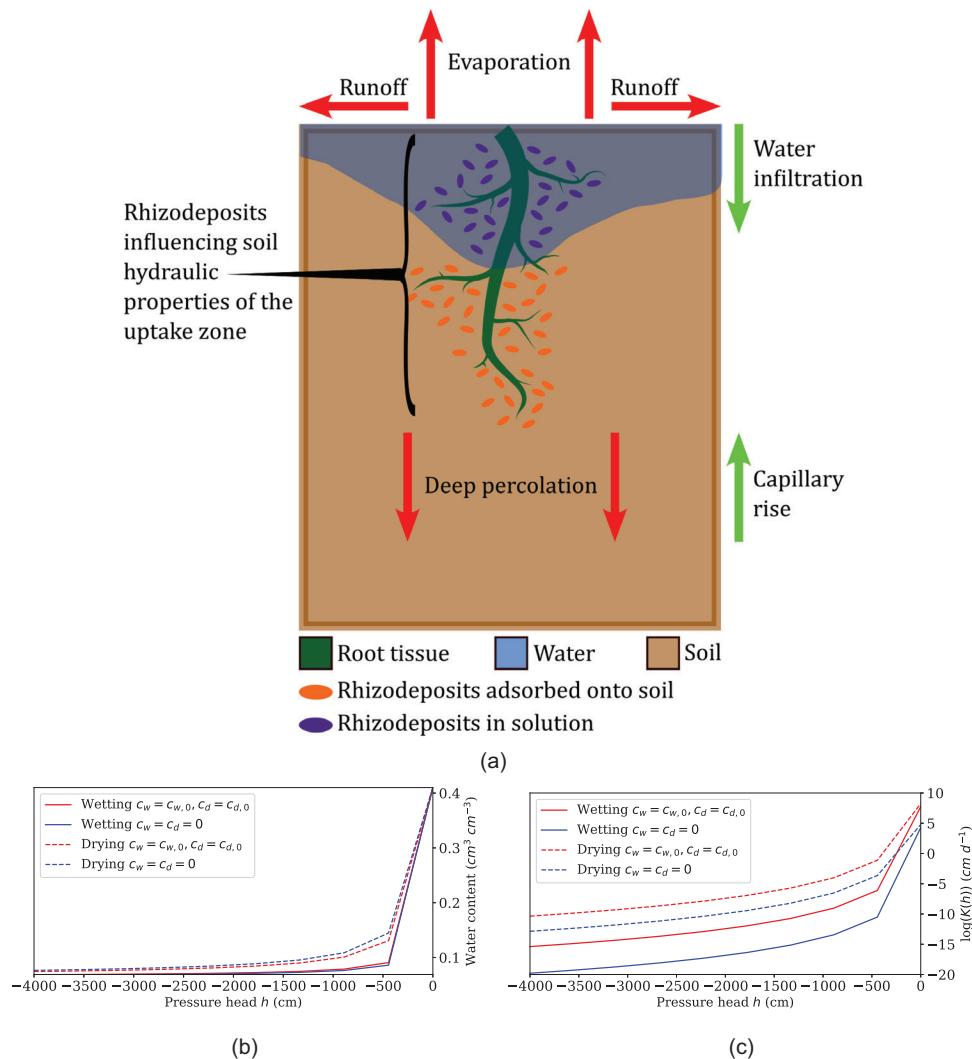


FIGURE 1 (a) Schematic representation of how rhizodeposits influence root water uptake. The uptake zone is the region of soil containing roots. Green arrows indicate the processes by which water enters this zone and red arrows represent the processes by which it is lost. By altering soil hydraulic properties, rhizodeposits affect these processes and hence influence the quantity of water available for uptake. (b) and (c) Graphs showing how the presence of rhizodeposits in solution c_w and rhizodeposits adsorbed to the soil c_d , at the initial concentrations used in simulations ($c_{d,0}$, $c_{w,0}$), affect (b) water retention θ and (c) hydraulic conductivity K for both wetting and drying soils.

to soil water surface tension and soil water contact angle. To consider these effects within rooted soil, Cooper et al. (2018) employed a multi-scale method to establish a formulation of Richards equation that incorporated rhizodeposit-induced changes to the viscosity and surface tension of soil water, as well as its contact angle with pore surfaces. The resulting model preserves microscale properties regarding the movement of air and water phases within soil pores (Cooper et al., 2017) but was not used to investigate how these rhizodeposit effects may influence root water uptake. Modeling studies where uptake has been investigated have taken a simpler approach by using one-dimensional Richards equations with water content and hydraulic conductivity functions whose parametrizations change according to radial distance from the root center (Carminati et al., 2010, 2011; Carminati, 2012).

Kroener et al. (2014), for example, shifted the pressure head term within the function for water content to account for the fraction of soil pore space occupied by mucilage, and the hydraulic conductivity incorporated the effect of mucilage-induced changes to viscosity of the soil water solution. Landl et al. (2021) have since coupled the model of Kroener et al. (2014) with a PDE for rhizodeposit dynamics to predict mucilage effects on local root water uptake.

Effectively predicting the impact of rhizodeposits on the total water uptake of a root system, however, requires models that can account for the full cycle of wet–dry–wet phases that occur over the entire soil depth during a given precipitation regime. Moreover, for each phase, these models should capture the potentially varying consequences on root water uptake of rhizodeposit-induced changes to soil hydraulic properties.

To address this, we propose a model composed of coupled differential equations for soil water transport and the dynamics of rhizodeposits in solution and rhizodeposits adsorbed to soil particles. The Richards equation was used for the soil water transport component of the model, with the parameters of the water content and hydraulic conductivity functions formulated so as to vary according to the rhizodeposit-induced changes to soil water surface tension, contact angle, and viscosity. Root surface area and length density functions were computed for generated RSAs of wheat (*Triticum aestivum* L.) and used to incorporate the distribution of rhizodeposit production throughout the soil, as well as parametrize a term for root water uptake that was formulated according to the model of Simunek and Hopmans (2009). The values of three model parameters for the effects of rhizodeposits on soil hydraulic properties were taken from the calibrated model of Gómez et al. (2025) in which the effect of rhizodeposits from wheat on soil water infiltration was accounted for, but (unlike the model we propose in this work) the physical presence of a root system taking up water and producing rhizodeposits was not incorporated. The remaining model parameter values are taken from existing literature.

The developed model is then used to investigate the hypothesis that, depending on the wetting and drying cycles during the precipitation regimes considered, the net effect of rhizodeposit presence on the total water uptake of a wheat root system may vary between being positive and negative. To this end, simulations from the model are carried out to examine the effect of rhizodeposit presence on the balance of water within the rooted zone during wetting and drying, while also assessing the resultant impact on total root water uptake. These simulations are carried out for six different 3-day environmental scenarios where there are two possibilities for the total quantity of precipitation, which can be delivered either over one, two or three rainfall events. Furthermore, for each environmental scenario the effect of rhizodeposits on root water uptake is simulated for three different wheat root systems: one that is 6 days old, one that is 15 days old, and one that is 30 days old.

2 | MATERIALS AND METHODS

2.1 | A coupled model for soil water and rhizodeposit dynamics

Assuming a soil domain with total depth L and spatial variable x , and a temporal interval with final time $T > 0$, a system of differential equations is used to model the relationship between soil hydraulics, root rhizodeposits, and root water uptake. As is standard in this field (Cooper et al., 2018; Kroener et al., 2014; Simunek et al., 2005), soil water transport is modeled using the Richards equation (L. A. Richards,

1931):

$$\begin{aligned} \partial_t \theta(h, c_w, c_d) + \partial_x q &= -S, & x \in (-L, 0), t \in (0, T], \\ q &= g_0, & x = 0, t \in (0, T], \\ q &= g_L, & x = -L, t \in (0, T], \\ h &= h_0, & x \in (-L, 0), t = 0. \end{aligned} \quad (1)$$

Here, (1) is solved for soil water pressure head h [L], with soil water content denoted by θ [$L^3 L^{-3}$], soil water flux by q [$L T^{-1}$], root water uptake by S [T^{-1}], and the flux conditions at the upper surface and base of the soil by g_0 and g_L respectively. At a given pressure head h , the influence on soil hydraulics by the concentrations of rhizodeposits in the soil solution c_w [$M L^{-3}$] and rhizodeposits adsorbed to the soil pore surface c_d [$M M^{-1}$] is incorporated in the water content function θ , and in the Darcy–Buckingham flux (Buckingham, 1907; Darcy, 1856):

$$q = -K(h, c_w, c_d) \partial_x (h + x), \quad (2)$$

where K [$L T^{-1}$] is the hydraulic conductivity. Similarly to Schnepf et al. (2012), the dynamics of c_w are modeled by an advection-diffusion equation

$$\begin{aligned} \partial_t (\theta c_w) - \partial_x (\theta D_w \partial_x c_w - q c_w) &= \rho \kappa_w c_d - \theta \kappa_d c_w + f, \\ x &\in (-L, 0), t \in (0, T], \\ \theta D_w \partial_x c_w - q c_w &= 0, & x = 0, -L, t \in (0, T], \\ c_w &= c_{w,0}, & x \in (-L, 0), t = 0, \end{aligned} \quad (3)$$

where the water flux q controls the advection of rhizodeposits in the soil solution. As in (1) and (2), the functions θ and q in (3) depend on h , c_w , and c_d but, for the sake of readability, these terms do not appear in their argument in (3). To capture the impedance to rhizodeposit diffusion caused by the tortuosity of the pore space, the coefficient $D_w = \phi D_{w,0}$ is used where ϕ is an impedance factor and $D_{w,0}$ is the value of the coefficient for free diffusion in water. Again, like in the work of Schnepf et al. (2012), the dynamics of rhizodeposits in solution (3) are coupled to an ordinary differential equation for the adsorbed concentration c_d via a first order reversible kinetic adsorption relation:

$$\begin{aligned} \rho \partial_t c_d &= \theta(h, c_w, c_d) \kappa_d c_w - \rho \kappa_w c_d, & x \in (-L, 0), t \in (0, T], \\ c_d &= c_{d,0}, & x \in (-L, 0), t = 0, \end{aligned} \quad (4)$$

where ρ [$M L^{-3}$] is the soil bulk density and κ_d [T^{-1}] and κ_w [T^{-1}] are the rates of rhizodeposit adsorption and desorption respectively. Unlike in the model of van de Weerd et al. (1999), the model in (4) assumes constant availability

of adsorption sites. Moreover, for the sake of more simple numerical simulation, we do not consider any changes to pore geometry that may occur from the accumulation of adsorbed rhizodeposits. The remaining term in (3) is then a source term f [$\text{M L}^{-3}\text{T}^{-1}$] for the release of rhizodeposits into the soil solution by the plant roots.

The influence of rhizodeposits on the hydraulic properties of soil is incorporated into the water transport and uptake model (1) through modifications to the van Genuchten (1980) and Mualem (1976) formulations of the functions for water content and hydraulic conductivity:

$$\theta(h, c_w, c_d) = \theta_r + (\theta_s - \theta_r)S_e(h, c_w, c_d), \quad (5)$$

$$K(h, c_w, c_d) = K^* + \alpha_K K_s(c_w, c_d) K_r(h, c_w, c_d). \quad (6)$$

Here, the effective soil saturation S_e in (5) and the saturated hydraulic conductivity K_s [L T^{-1}] and effective hydraulic conductivity K_r in (6) are functions of pressure head and rhizodeposit concentration. The terms θ_r and θ_s in (5) denote the residual and saturated soil water contents, respectively, and parameters K^* [L T^{-1}] and α_K are included in (6) to account for the fact that water content and hydraulic conductivity values at a given pressure head are lower if a soil is wetting than if it is drying, see (12). This hysteresis exists because the processes that occur when a soil drains are generally governed by activity in smaller pores, whereas during wetting it is the activity within larger pores that is more consequential. Also, the contact angles between the menisci and the surface of soil particles are different depending on whether water is entering or draining from a pore (van Genuchten & Pachepsky, 2011; Zhou, 2013). In (5) and (6), hysteresis is incorporated in the effective saturation

$$S_e(h, c_w, c_d) = \frac{1}{(1 + |\alpha(c_w, c_d)h|^n)^m}, \quad (7)$$

through the function α , see (9). The resulting expression for the effective hydraulic conductivity in (6) is as follows:

$$K_r(h, c_w, c_d) = \left(\frac{2(\theta_{s,0} - \theta_{r,0})S_e(h, c_w, c_d) + \varepsilon(1 - 2S_e(h, c_w, c_d))}{2(\theta_{s,0} - \theta_{r,0})} \right)^l \times \left[1 - \left(1 - \left(\frac{2(\theta_{s,0} - \theta_{r,0})S_e(h, c_w, c_d) + \varepsilon(1 - 2S_e(h, c_w, c_d))}{2(\theta_{s,0} - \theta_{r,0})} \right)^{\frac{1}{m}} \right)^m \right]^2, \quad (8)$$

where n and $m = 1 - \frac{1}{n}$ are shape parameters, l denotes the tortuosity of the porous structure (Mualem, 1976; van Genuchten, 1980) and $\varepsilon > 0$ is a small regularization parameter whose inclusion allows Equations (1)–(4) to be solved numerically (List & Radu, 2016; Mair et al., 2022).

For a depth x and time t the function α [L^{-1}] takes different values depending on whether the soil is wetting (increasing h) or drying (decreasing h):

$$\alpha(x, t, c_w, c_d) = \begin{cases} \alpha^w(c_w, c_d) & \text{if the soil at depth } x \text{ is wetting at time } t, \\ \alpha^d(c_w, c_d) & \text{if the soil at depth } x \text{ is drying at time } t, \end{cases} \quad (9)$$

with $\alpha^w(c_w, c_d) \geq \alpha^d(c_w, c_d)$ for any nonnegative (c_w, c_d) (Kool & Parker, 1987). Similarly, in (6) hysteresis is also incorporated through the saturated hydraulic conductivity where:

$$K_s(x, t, c_w, c_d) = \begin{cases} K_s^w(c_w, c_d) & \text{if the soil at depth } x \text{ is wetting at time } t, \\ K_s^d(c_w, c_d) & \text{if the soil at depth } x \text{ is drying at time } t, \end{cases} \quad (10)$$

and $K_s^d(c_w, c_d) > K_s^w(c_w, c_d)$ for any nonnegative (c_w, c_d) (Vogel et al., 1996). If at some depth and time (x_Δ, t_Δ) the soil switches from wetting to drying, then $\alpha^w(c_w, c_d)$ and $K_s^w(c_w, c_d)$ switch to $\alpha^d(c_w, c_d)$ and $K_s^d(c_w, c_d)$ accordingly, or vice versa if the switch is from drying to wetting. To maintain the continuity of θ and K under transitions between wetting and drying, some other parameters in (5) and (6) must also change values. Specifically, with the pressure head, water content, hydraulic conductivity, and rhizodeposit concentrations at the switch point (x_Δ, t_Δ) denoted as h_Δ , θ_Δ , K_Δ , $c_{w,\Delta}$, and $c_{d,\Delta}$, respectively, the residual and saturated water contents assume the values

$$\theta_r(x_\Delta, t_\Delta) = \begin{cases} \frac{\theta_{\Delta} - \theta_{s,0} S_e(h_{\Delta}, c_{w,\Delta}, c_{d,\Delta})}{1 - S_e(h_{\Delta}, c_{w,\Delta}, c_{d,\Delta})} & \text{if wetting at } (x_\Delta, t_\Delta), \\ \theta_{r,0} & \text{if drying at } (x_\Delta, t_\Delta), \end{cases}$$

$$\theta_s(x_\Delta, t_\Delta) = \begin{cases} \theta_{s,0} & \text{if wetting at } (x_\Delta, t_\Delta), \\ \frac{\theta_{\Delta} - \theta_{r,0}(1 - S_e(h_{\Delta}, c_{w,\Delta}, c_{d,\Delta}))}{S_e(h_{\Delta}, c_{w,\Delta}, c_{d,\Delta})} & \text{if drying at } (x_\Delta, t_\Delta), \end{cases} \quad (11)$$

where $\theta_{r,0}$ and $\theta_{s,0}$ are the original values of residual and saturated soil water content. Moreover, with the wetting and drying saturated hydraulic conductivities at (x_Δ, t_Δ) denoted by $K_{s,\Delta}^w = K_s^w(c_{w,\Delta}, c_{d,\Delta})$ and $K_{s,\Delta}^d = K_s^d(c_{w,\Delta}, c_{d,\Delta})$, respectively, parameters K^* and α_K take the following values (Vogel et al., 1996):

$$K^*(x_\Delta, t_\Delta) = \begin{cases} K_{s,\Delta}^w K_r(0) \left(1 - \frac{K_{\Delta} - K_{s,\Delta}^w K_r(0)}{K_{s,\Delta}^w K_r(h_{\Delta}, c_{w,\Delta}, c_{d,\Delta}) - K_{s,\Delta}^w K_r(0)} \right) & \text{if wetting at } (x_\Delta, t_\Delta), \\ 0 & \text{if drying at } (x_\Delta, t_\Delta), \end{cases} \quad (12)$$

$$\alpha_K(x_\Delta, t_\Delta) = \begin{cases} \frac{K_{\Delta} - K_{s,\Delta}^w K_r(0)}{K_{s,\Delta}^w K_r(h_{\Delta}, c_{w,\Delta}, c_{d,\Delta}) - K_{s,\Delta}^w K_r(0)} & \text{if wetting at } (x_\Delta, t_\Delta), \\ \frac{K_{\Delta}}{K_{s,\Delta}^d K_r(h_{\Delta}, c_{w,\Delta}, c_{d,\Delta})} & \text{if drying at } (x_\Delta, t_\Delta), \end{cases}$$

where $c_{w,\Delta}$ and $c_{d,\Delta}$ are omitted from the argument of K_r at $h = 0$ since they have no effect on its value once the soil is fully saturated. At x_Δ , these new values, (11) and (12), are then maintained until the next time that the soil at the given depth x_Δ switches from wetting to drying (or vice versa), when they then have to be updated to new values using the same formulae. The explicit dependence on space and time that is imposed on θ , K , S_e , K_r (and their respective parameters) through these hysteresis formulae applies within every component of the developed model system, but, in the interest of readability, the variables x and t are shown only in the arguments of the functions in formulae (9)–(12).

The surface tension γ [M T⁻²] and viscosity μ [M L T⁻¹] of soil water, along with the contact angle ϑ of its menisci with pore surfaces, have been shown to vary considerably according to the abundance of rhizodeposits, and the species of plant from which these were released (Ahmed et al., 2016; Naveed et al., 2018, 2019). The analysis here focuses on wheat plant roots (*T. aestivum* L.) where there is evidence that when the rhizodeposits they release are mixed in the soil water, they decrease its surface tension (Gómez et al., 2025; Read et al., 2003) and increase its viscosity (Hosseini and Mosaddeghi, 2024), and when they are adsorbed to pore surfaces, they increase the soil water contact angle (Zickenrott et al.,

2016). As a result, the approach of Karagunduz et al. (2001) is employed to account for these effects by introducing scaling factors into α , such that for $(c_w, c_d) \in [0, \infty)^2$,

$$\alpha^w(c_w, c_d) = \frac{\gamma(c_w) \cos(\vartheta_0)}{\gamma_0 \cos(\vartheta(c_d))} \alpha_0^w \quad \text{and} \quad \alpha^d(c_w, c_d) = \frac{\gamma_0 \cos(\vartheta(c_d))}{\gamma(c_w) \cos(\vartheta_0)} \alpha_0^d, \quad (13)$$

where γ_0 and ϑ_0 are the surface tension and contact angle values in the absence of rhizodeposits, and α_0^w and α_0^d are the corresponding wetting and drying values of α . To further incorporate into K the effect of a rhizodeposit-induced increase in the soil water's viscosity, decrease in its surface tension, and increase in its contact angle with the pore surface,

the saturated hydraulic conductivity is such that, for $(c_w, c_d) \in [0, \infty)^2$,

$$K_s^z(c_w, c_d) = K_{s,0}^z \left(\frac{\gamma_0}{\gamma(c_w)} \right)^\beta \frac{\cos(\vartheta(c_d)) \mu_0}{\cos(\vartheta_0) \mu(c_w)}, \quad (14)$$

where $z = w, d$ and μ_0 is water viscosity in the absence of rhizodeposits. This formulation of saturated hydraulic conductivity (14) was informed by the data of Gómez et al. (2025) where it was found that soil wetting was facilitated by the presence of rhizodeposits of wheat (*T. aestivum* L. var. Filon) in the soil solution. Moreover, this facilitation outweighed the soil hydrophobicity caused when the rhizodeposits had been dried onto the particles prior to the experiment. Assuming that a rhizodeposit-induced reduction in liquid surface tension was what caused the accelerated soil wetting, the exponent β in (14) determined the extent to which the infiltration facilitation from reduced surface tension dominated over the effects of any reductions to hydraulic conductivity from rhizodeposits in solution increasing viscosity or adsorbed rhizodeposits increasing particle hydrophobicity with higher soil water contact angles.

With the relations formulated in (13) and (14), the functions for soil water content θ in (5) and hydraulic conductivity K in (6) then capture the mechanisms by which

rhizodeposit-induced changes to soil hydraulic properties affect the balance of forces that govern water retention and flow. In the formulations of (13) a rhizodeposit-induced reduction in surface tension will decrease α^w , which means that during wetting the water content at a given pressure head will be increased, thus also increasing the hydraulic conductivity at that pressure head. According to (14), hydraulic conductivity will also be directly increased by reduced surface tension through an increase to the values of K_s^w and K_s^d . For a given region of soil, these effects on water retention and hydraulic conductivity mean that reductions to surface tension will facilitate the flow of water both into and out of the region, according to pressure gradients and the effects of gravity (2). This effect of surface tension is countered by the effects of rhizodeposit-induced increases to both the viscosity of the soil water solution and the contact angle that its menisci make with the pore surface. Increases to solution viscosity reduce the hydraulic conductivity of the soil (Xiao et al., 2024), which slows the infiltration caused by pressure gradients and gravity, thus increasing the retention of water in areas of the soil with higher water contents. The increases to contact angle caused by adsorbed rhizodeposits similarly impedes the movement of water from wetter to drier regions of soil, but by increasing the water repellency of the pore surfaces in the drier soil as opposed to altering any properties of the solution in the wetter region (Ahmed et al., 2016). Figure 1b,c shows the effect of these changes to surface tension, viscosity, and contact angle on water retention (b) and hydraulic conductivity (c) when the concentrations of rhizodeposits in solution and rhizodeposits adsorbed to the pore surfaces are set to the initial conditions used in all the simulations that were performed.

In addition to altering soil hydraulic properties through rhizodeposition, it has also been shown that by soil penetration and particle rearrangement (Angers & Caron, 1998; Feeney et al., 2006), plant roots can create macropore channels along their growth trajectories (Noguchi et al., 1999). The combination of these effects has since been observed to induce preferential flows of water through the soil (Ghestem et al., 2011; Luo et al., 2019), which are oriented according to the architecture of the root system present, and thus exert a considerable influence on the path soil water takes during post-precipitation infiltration (Noguchi et al., 1997). This phenomenon of preferential flow has previously been incorporated in models for soil water transport (Gerke & van Genuchten, 1993; Mair et al., 2022) and its consequences on the availability of water for uptake evaluated (Mair et al., 2023). However, when considering depth as the only spatial dimension of a model for soil water transport, as in (1), the lateral architectural features of a root system and the channels they create are not seen, and preferential flow equates to facilitated infiltration in the vertical direction. The omission of any direct incorporation of preferential flow in model (1) is therefore justified since it already incorporates

rhizodeposit-induced changes to soil hydraulic properties, including facilitation to infiltration, and, as explained, this has itself been established as a causal factor behind preferential flow (Mair et al., 2023). Were a three-dimensional version of model (1) to be considered, then the omission of preferential flow would be more difficult to justify.

Root water uptake in (1) is modeled with a macroscopic sink term S incorporating water availability, root distribution, and potential plant transpiration $\mathcal{T}_p$ [L T⁻¹] (Simunek and Hopmans, 2009):

$$S(x, t, h) = \alpha_f(h) \text{NRLD}(x) \mathcal{T}_p \frac{1}{\max(\omega(t), \omega_c)}. \quad (15)$$

The dependence of uptake on the level of soil water content is included in S through the dimensionless stress response function $0 \leq \alpha_f \leq 1$, and the plant's capacity to reach the water within the soil is accounted for by the normalized root length density NRLD [L⁻¹], which is a continuous function that gives the distribution of root length within the soil. The potential plant transpiration is formulated as $\mathcal{T}_p = \text{ET}_0 K_{cb}$, where ET_0 [L T⁻¹] is the reference evapotranspiration and K_{cb} [–] is the basal crop coefficient (Allen et al., 1998). The measure of the global water stress experienced by the plant is

$$\omega(t) = \int_{\Omega} \alpha_f(h(x, t)) \text{NRLD}(x) dx, \quad (16)$$

and the capacity of the plant to compensate for poor water availability in one soil region by increasing uptake in zones of higher saturation is reflected in the value of the critical water stress index ω_c (Cai et al., 2018; Simunek and Hopmans, 2009). The boundary condition at the soil surface is

$$g_0(h, t) = K_e(h) \text{ET}_0 + \text{RO}(h, t) - P(t), \quad (17)$$

where K_e [–] is a function that uses the soil water content to determine the proportion of total evapotranspiration that comes from evaporation (Allen et al., 1998). A full formulation of K_e is given in the supplementary material to the work of Mair et al. (2023) and the values of the function parameters that we used for the simulations in this article are displayed in Table 1. In the interest of avoiding formulating a model that would be too computationally expensive to simulate with the resources available, model (1) and the boundary condition (17) do not account for any dependence of atmospheric water dynamics on belowground processes. Specifically, ET_0 [L T⁻¹] was set to a constant value with no dependence on soil water dynamics (Allen et al., 1998, Chapter 1, Table 2).

The precipitation rate in (17) is given by P [L T⁻¹] and RO [L T⁻¹] is the run-off of water that occurs when precipitation cannot enter the soil due to the surface already being

TABLE 1 Notation and definition of terms that appear in this article.

Term	Description	Value	Reference
a	Steepness of transition to complete run off once the boundary approaches saturation	250	n/a
c_d	Concentration of rhizodeposits adsorbed to soil surface (mg mg^{-1})	n/a	n/a
$c_{d,0}$	Initial concentration of rhizodeposits adsorbed to soil surface (mg mg^{-1})	n/a	n/a
$c_{d,\Delta}$	Concentration of rhizodeposits adsorbed to soil surface at hysteresis reversal point (mg mg^{-1})	n/a	n/a
c_w	Concentration of rhizodeposits in solution (mg cm^{-3})	n/a	n/a
$c_{w,0}$	Initial concentration of rhizodeposits in solution (mg cm^{-3})	n/a	n/a
$c_{w,\Delta}$	Concentration of rhizodeposits in solution at hysteresis reversal point (mg cm^{-3})	n/a	n/a
D_w	Rhizodeposit diffusion coefficient ($\text{cm}^2 \text{d}^{-1}$)	0.65	(Scott et al., 1995)
ET_0	Reference evapotranspiration (cm d^{-1})	0.1	(Allen et al., 1998)
f	Rhizodeposit release rate ($\text{mg cm}^{-3} \text{d}^{-1}$)	n/a	n/a
f_c	Fraction of soil surface covered by vegetation (-) (appears in K_e)	0.1	(Allen et al., 1998, Chap. 7, tab. 21)
f_w	Fraction of soil surface wetted by precipitation (-) (appears in K_e)	1.0	(Allen et al., 1998, Chap. 7, tab. 20)
g_0	Water flux at upper soil surface (cm d^{-1})	n/a	n/a
g_L	Water flux at lower soil surface (cm d^{-1})	n/a	n/a
h	Soil water pressure head (cm)	n/a	n/a
h_0	Initial pressure head condition (cm)	n/a	n/a
h_1	Pressure head associated with anaerobiosis due to soil saturation (cm) (appears in α_f)	0	(Wesseling, 1991, tab. 5)
h_2	Upper limit of pressure head interval in which uptake capacity is maximized (cm) (appears in α_f)	-1.0	(Wesseling, 1991, tab. 5)
h_3	Lower limit of pressure head interval in which uptake capacity is maximized (cm) (appears in α_f)	-500.0	(Wesseling, 1991, tab. 5)
h_4	Pressure head associated with plant wilting point (cm) (appears in α_f)	-16000.0	(Wesseling, 1991, tab. 5)
h_Δ	Pressure head at hysteresis reversal point (cm)	n/a	(Kool & Parker, 1987)
K	Hydraulic conductivity (cm d^{-1})	n/a	(Mualem, 1976; van Genuchten, 1980)
K_e	Evaporation coefficient (-)	n/a	(Allen et al., 1998; Mair et al., 2023)
K_r	Effective hydraulic conductivity (-)	n/a	(Mualem, 1976; van Genuchten, 1980)
K_s	Hysteretic saturated hydraulic conductivity (cm d^{-1})	n/a	(Vogel et al., 1996)
K_s^d	Saturated hydraulic conductivity of drying hydraulic conductivity curve (cm d^{-1})	n/a	(Vogel et al., 1996)
$K_{s,0}^d$	Saturated hydraulic conductivity of drying hydraulic conductivity curve in absence of rhizodeposits (cm d^{-1})	120	(Vogel et al., 1996)
$K_{s,\Delta}^d$	Saturated hydraulic conductivity of drying hydraulic conductivity curve at hysteresis reversal point (cm d^{-1})	n/a	n/a
K_s^w	Saturated hydraulic conductivity of wetting hydraulic conductivity curve (cm d^{-1})	n/a	(Vogel et al., 1996)
$K_{s,0}^w$	Saturated hydraulic conductivity of wetting hydraulic conductivity curve in absence of rhizodeposits (cm d^{-1})	72	(Vogel et al., 1996)
$K_{s,\Delta}^w$	Saturated hydraulic conductivity of wetting hydraulic conductivity curve at hysteresis reversal point (cm d^{-1})	n/a	n/a
K_Δ	Hydraulic conductivity at hysteresis reversal point (cm d^{-1})	n/a	(Vogel et al., 1996)
K^*	Fictitious hysteresis parameter in hydraulic conductivity (cm d^{-1})	n/a	(Vogel et al., 1996)
K_{cb}	Basal crop coefficient (-)	n/a	(Allen et al., 1998, Chap. 7, tab. 17)

(Continues)

TABLE 1 (Continued)

Term	Description	Value	Reference
K_{cmax}	Maximum crop coefficient (-) (appears in K_e)	1.30	(Allen et al., 1998, Cha. 7, tab. 19)
l	Pore tortuosity parameter in hydraulic conductivity function (-)	0.5	(van Genuchten & Pachepsky, 2011)
L	Soil depth (cm)	53	n/a
[L]	Length dimension	n/a	n/a
m	Pore size parameter of soil water retention curve	0.47	(Mualem, 1976; van Genuchten, 1980)
[M]	Mass dimension	n/a	n/a
n	Pore size parameter of soil water retention curve	1.89	(Mualem, 1976; van Genuchten, 1980)
NRLD	Normalized root length density (cm^{-1})	n/a	(Mair et al., 2023)
P	Precipitation rate (cm d^{-1})	n/a	n/a
q	Soil water flux (cm d^{-1})	n/a	n/a
RO	Surface runoff (cm d^{-1})	n/a	n/a
S	Root water uptake rate (cm d^{-1})	n/a	(Simunek & Hopmans, 2009)
S_e	Effective saturation (-)	n/a	(Mualem, 1976; van Genuchten, 1980)
SAD	Root surface area density (cm^{-1})	n/a	(Mair et al., 2023)
t	Temporal variable (d)	n/a	n/a
t_{Δ}	Time of hypothetical hysteresis reversal point (d)	n/a	n/a
T	Final time (d)	3	n/a
[T]	Time dimension	n/a	n/a
$\mathcal{T}_p$	Potential plant transpiration (cm d^{-1})	n/a	(Allen et al., 1998)
x	Spatial variable (cm)	n/a	n/a
x_{Δ}	Position of hypothetical hysteresis reversal point (cm)	n/a	n/a
α	Shape parameter of soil water retention curve (cm^{-1})	n/a	(Kool & Parker, 1987)
α_f	Water stress response (-)	n/a	(Simunek and Hopmans, 2009)
α_K	Hydraulic conductivity hysteresis scaling parameter (cm d^{-1})	n/a	(Vogel et al., 1996)
α^d	Value of α in drying branch of soil water retention curve (cm^{-1})	n/a	(Kool & Parker, 1987)
α_0^d	Value of α in drying soil water retention curve in absence of rhizodeposits (cm d^{-1})	0.0114	(Carsel & Parrish, 1988)
α^w	Value of α in wetting branch of soil water retention curve (cm^{-1})	n/a	(Kool & Parker, 1987)
α_0^w	Value of α in wetting soil water retention curve in absence of rhizodeposits (cm d^{-1})	0.0521	(Carsel & Parrish, 1988)
β	Fitting parameter in K_s^w and K_s^d (-)	8.772	(Gómez et al., 2025)
ϵ	Regularization parameter in effective hydraulic conductivity (-)	10^{-15}	n/a
γ	Surface tension of soil water (mN m^{-1})	n/a	n/a
γ_0	Value of γ in absence of rhizodeposits (mN m^{-1})	72.9	(Read et al., 2003)
κ_d	Rhizodeposit drying rate (d^{-1})	9.94×10^{-2}	(Gómez et al., 2025)
κ_w	Rhizodeposit wetting rate (d^{-1})	1.78×10^{-2}	(Gómez et al., 2025)
λ	Rate of rhizodeposit release per unit root surface area ($\text{mg cm}^{-2} \text{d}^{-1}$)	0.039	(Ptashnyk et al., 2011)
μ	Viscosity of soil water (mPa s)	n/a	n/a
μ_0	Value of μ in absence of rhizodeposits (mPa s)	0.89	(Hosseini & Mosaddeghi, 2024)
ϕ	Impedance factor (-)	0.3	(Schnepf et al., 2012)

(Continues)

TABLE 1 (Continued)

Term	Description	Value	Reference
ρ	Soil bulk density (mg cm^{-3})	1650	(Morris & Lowery, 1988)
θ	Soil water content ($\text{cm}^3 \text{cm}^{-3}$)	n/a	(Mualem, 1976; van Genuchten, 1980)
θ_{fc}	Water content at field capacity ($\text{cm}^3 \text{cm}^{-3}$) (appears in K_e)	0.28	(Allen et al., 1998, Chap. 7, tab. 19)
θ_r	Hysteretic residual water content ($\text{cm}^3 \text{cm}^{-3}$)	n/a	(Kool & Parker, 1987)
$\theta_{r,0}$	Non-hysteretic residual water content ($\text{cm}^3 \text{cm}^{-3}$)	0.065	(Carsel & Parrish, 1988)
θ_s	Hysteretic saturated water content ($\text{cm}^3 \text{cm}^{-3}$)	n/a	(Kool & Parker, 1987)
$\theta_{s,0}$	Non-hysteretic saturated water content ($\text{cm}^3 \text{cm}^{-3}$)	0.41	(Carsel & Parrish, 1988)
θ_{wp}	Water content at wilting point ($\text{cm}^3 \text{cm}^{-3}$) (appears in K_e)	0.16	(Allen et al., 1998, Chap. 7, tab. 19)
θ_{Δ}	Water content at hysteresis reversal point ($\text{cm}^3 \text{cm}^{-3}$)	n/a	(Kool & Parker, 1987)
ϑ	Contact angle between soil water and pore surface (rad)	n/a	n/a
ϑ_0	Value of ϑ in absence of rhizodeposits (rad)	0.0294	(Zickenrott et al., 2016)
ω	Global water stress (-)	n/a	(Simunek & Hopmans, 2009)
ω_c	Critical water stress index (-)	0.8	(Cai et al., 2018)

Note: The value column reads n/a if the corresponding term is a variable, a function, or a parameter that is altered between simulations. References included indicate if the parameter value or function formulation were taken from a specific source.

fully saturated:

$$\text{RO}(h, t) = P(t) \left(1 + \frac{\exp(a(\theta(h) - \theta_{s,0})) - 1}{\exp(a(\theta(h) - \theta_{s,0})) + 1} \right). \quad (18)$$

Here, the constant $a > 0$ is set to a large enough value so that if the surface is fully saturated ($\theta = \theta_{s,0}$) then $\text{RO} = P$.

On the lower boundary $x = -L$, the condition of free drainage is assumed, which translates as $g_L(h) = -K(h, c_w, c_d)$. Finally, the source term f in (3), for the release of rhizodeposits into the soil water solution, is formulated as follows:

$$f(x, h) = \lambda \theta(h) \text{SAD}(x), \quad (19)$$

where $\lambda [\text{ML}^{-2}\text{T}^{-1}]$ is the rate at which rhizodeposits are released from the surface of the root (Ptashnyk et al., 2011). The term $\text{SAD} [\text{L}^2\text{L}^{-3}]$ is the root surface area density, a continuous function of space representing the distribution of root surface area throughout the soil, and this multiplies λ to determine the rate of release in each region of the soil. The factor of θ in (19) incorporates the assumption that the concentration c_w of rhizodeposits in the soil water can only increase in regions of the soil where there is water to which rhizodeposits can be added.

2.2 | Parametrization and calibration

The functions for normalized root length density NRLD and surface area density SAD were constructed for three wheat

root systems of ages 6, 15, and 30 days. These root systems were generated from CRootBox (Schnepf et al., 2018) using the default settings for wheat systems. The respective RSAs are shown in Figure 2. The methods previously employed and described by Mair et al. (2022, 2023) were used to convert the architecture data of each root system into density functions for the distributions of root length and surface area. These functions took values in a three-dimensional soil domain defined by the furthest reaching roots across each of the root systems, and were then integrated across the lateral dimensions to produce NRLD and SAD as functions of soil depth ($-L, 0$). Here $L = 53$ cm is the greatest depth reached by any root of the three systems. To explicitly model the ways in which rhizodeposits induce an increase in soil water viscosity μ , a reduction in its surface tension γ , and an increase in the contact angle ϑ between menisci and the soil pore surface, the data of Hosseini and Mosaddeghi (2024), Read et al. (2003), and Zickenrott et al. (2016) were used to obtain the following functions:

$$\mu(c_w) = 0.89 + 0.09(1 + \exp(-6.01(c_w + 0.16)))^{-10}, \quad (20)$$

$$\gamma(c_w) = 47.5 + \frac{72.9 - 47.5}{1 + \exp(18.79(c_w - 0.36))}, \quad (21)$$

$$\vartheta(c_d) = \frac{3.53}{18} \pi (1 + \exp(-1879.14c_d))^{-4.39}. \quad (22)$$

These functions return $\mu_0 = \mu(0) = 0.89$ mPa · s, $\gamma_0 = \gamma(0) = 72.9$ mN m⁻¹, and $\vartheta_0 = \vartheta(0) = 0.0294$ rad as the viscosity, surface tension, and contact angle when there are no

FIGURE 2 Diagram of pipeline for simulating each of the different precipitation scenarios for each root system and investigating the effect of rhizodeposits on root water uptake.

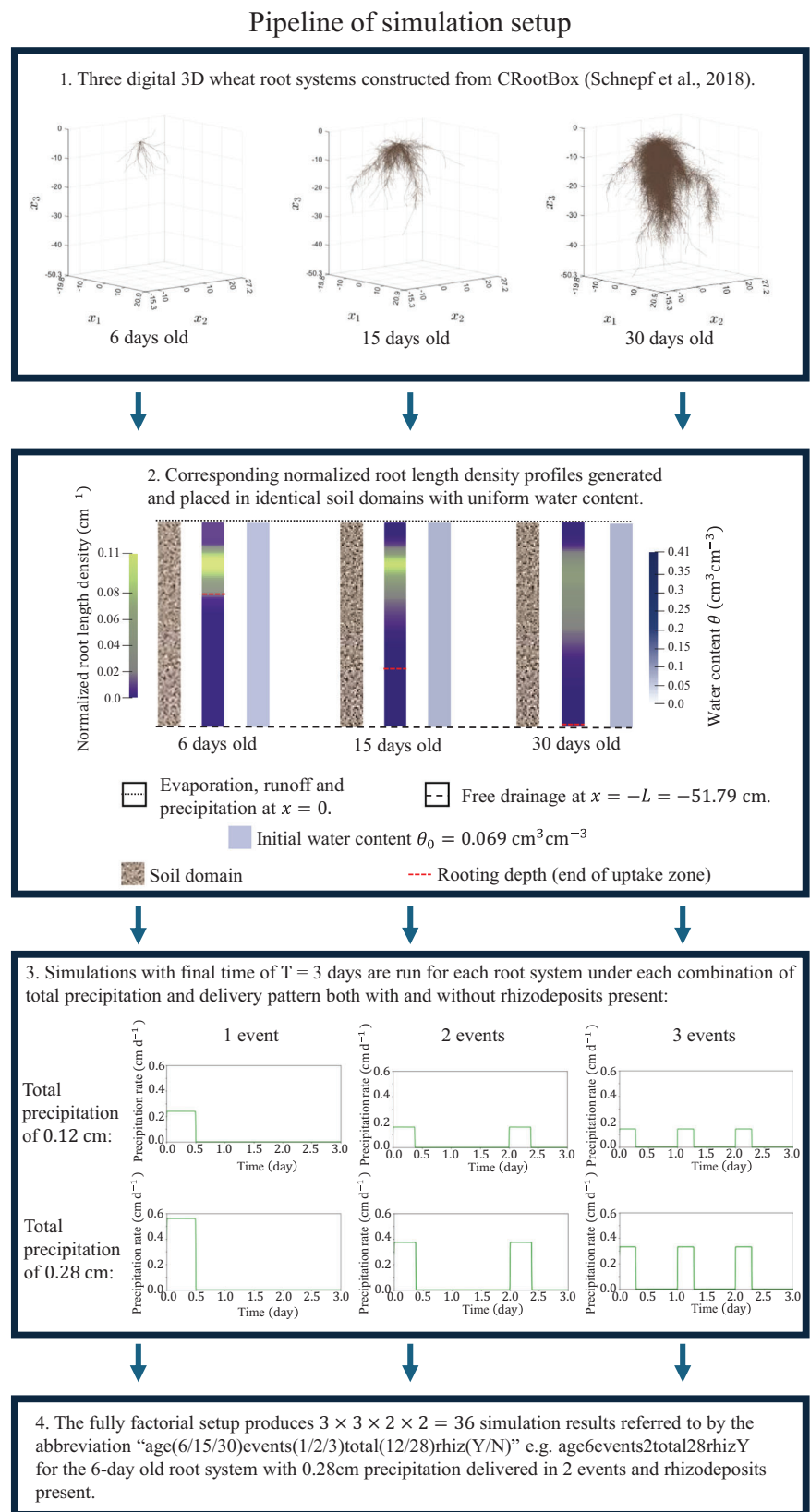


TABLE 2 List of abbreviations that are used within this article.

Abbreviation	Meaning
age(6/15/30)events(1/2/3)total(12/28)	Used to refer to a pair of simulated scenarios, one with rhizodeposits present and the other without, where the conditions of root system age (6, 15, or 30 days old), total amount of precipitation (0.12 or 0.28 cm), and rainfall delivery pattern (one, two, or three events) are fixed. As an example, age6events2total28 refers to the pair of simulations, one with rhizodeposits present and the other without where, in both cases, the root system is 6 days old and a total precipitation of 0.28 cm is split over two rainfall events.
age(6/15/30)events(1/2/3)total(12/28)rhiz(Y/N)	Used to refer to a specific simulated scenario where the root system is either 6, 15, or 30 days old, a total amount of precipitation of either 0.12 or 0.28 cm is delivered in either one, two, or three rainfall events, and rhizodeposits are either present or absent. As an example, age6events2total28rhizY is the abbreviation for the scenario where the root system is 6 days old, the delivery of a total precipitation of 0.28 cm is split over two rainfall events, and rhizodeposits are present in the soil.
age(6/15/30)total(12/28)	Used to refer to a group of six simulated scenarios where the conditions of root system age (6, 15, or 30 days old) and total amount of precipitation (0.12 or 0.28 cm) are fixed, but, between simulations, the conditions of precipitation delivery pattern (one, two, or three events) and rhizodeposit presence/absence are varied. As an example, age6total28 refers to the six simulations that all consider a 6-day old root system with total precipitation set to 0.28 cm, but between them the conditions of precipitation delivery pattern and rhizodeposit presence/absence vary.
events(1/2/3)total(12/28)	Used to refer to a group of six simulated scenarios where the conditions for the total amount of precipitation (0.12 or 0.28 cm) and the pattern by which it is delivered (one, two, or three events) are fixed, but, between simulations, the conditions of root system age (6, 15, or 30 days) and rhizodeposit presence/absence are varied. As an example, events2total28 refers to the six simulations that all consider a total precipitation of 0.28 cm delivered over two rainfall events, but between them the conditions for age of root system and rhizodeposit presence/absence vary.
events(1/2/3)total(12/28)rhiz(Y/N)	Used to refer to a group of three simulated scenarios where the conditions for total amount of precipitation (0.12 or 0.28 cm), the pattern by which precipitation is delivered (one, two, or three events), and rhizodeposit presence/absence are fixed, but, between simulations, the choice of root system age (6, 15, or 30 days) is varied. As an example, events2total28rhizY refers to the three simulations that all consider a total precipitation of 0.28 cm that is delivered over two rainfall events with rhizodeposits present, but between them the age of root system considered varies between 6, 15, or 30 days old.
NRLD	Normalized root length density
PDE	Partial differential equation
RSA	Root system architecture
SAD	Root surface area density

rhizodeposits present. The precipitation term P is formulated as a step function of time t so that within chosen intervals of the simulation it prescribes constant rainfall at a given rate but outside these it takes the value of 0 cm. A full description of the formulation of these type of functions is available in the supplementary material that accompanies the work of Mair et al. (2023).

A sandy loam soil was considered in all model simulations, and the majority of the values required to parametrize (1)–(4) accordingly were taken from existing literature, see Table 1. Gómez et al. (2025) recently employed a model similar to (1)–(4), however it did not incorporate the effects of a root system taking up water and producing rhizodeposits. By calibrating against their own experimental data, Gómez et al. (2025) obtained the values of parameters in their model for the desorption rate of adsorbed rhizodeposits κ_w , the rate at which rhizodeposits in solution adsorb to the pore surface κ_d , and β

the extent to which a rhizodeposit-induced reduction in surface tension increases hydraulic conductivity. This data came from dye tracing experiments, in which chambers were filled with three layers of the artificial transparent soil Nafion, and the infiltration of dye solution through the medium was monitored. They added extracted wheat rhizodeposits to the dye solution in the upper Nafion layer and the effect on infiltration was recorded. An advection-diffusion equation for dye concentration within the chamber was used by Gómez et al. (2025) to simulate this experimental process, and they employed Bayesian optimization (Brochu et al., 2010) to approximate the values $(\kappa_d, \kappa_w, \beta)$ that provided outputs from their model that most accurately matched their experimental results. Due to the absence of alternative options, the parameter values identified by their calibration are now also used in this work to parametrize model (1)–(4), and are cited in Table 1. This parametrization choice implies the assumption

that the values of parameters $(\kappa_d, \kappa_w, \beta)$ for Nafion can also be applied when considering sandy loam soils. Regarding the values of (κ_d, κ_w) , since the properties of Nafion differ from those of natural soils, especially in terms of particle heterogeneity, it is clear that this assumption is not optimal. Theoretically, however, the value of β only describes the extent of the correction that should be made to the hydraulic conductivity of a given soil when accounting for the presence of rhizodeposits and, therefore, its value may exhibit less of a change between different soil types. Further space in the discussion section is dedicated to the acknowledgment and consideration of the potential consequences of this parametrization choice.

2.3 | Simulated scenarios

The simulation pipeline that was followed to investigate the impact of rhizodeposits on root water uptake can be seen in Figure 2. By using the NRLD and SAD functions of the 6-, 15-, and 30-day old root systems, three parametrizations of model system (1)–(4) were obtained, one for each root system age. Additionally, using data for the initial, mid-season and end of season values of the basal crop coefficient of wheat (Allen et al., 1998, Chapter 7, Table 17), a piecewise linear function of K_{cb} with respect to plant age was formulated and this resulted in values of $K_{cb} = 0.15, 0.32, 0.5$ within the uptake functions corresponding to the 6-, 15-, and 30-day old root systems, respectively. For each of the three root systems considered, and with a final time of $T = 3$ days, simulations of the model system (1)–(4) were then run with the precipitation function P imposing a total precipitation of either 0.12 or 0.28 cm and delivering this rainfall via either one, two, or three events. In all simulations it was assumed that root systems were not growing.

The fully factorial simulation program that combined every precipitation regime with every root system both with and without rhizodeposits resulted in a total of 36 simulations with each being referred to by the abbreviation `age(6/15/30)events(1/2/3)total(12/28)rhiz(Y/N)`, for example, `age6events2total28rhizY` for the simulation with a 6-day old root system, rhizodeposits present, and a precipitation regime where 0.28 cm total rainfall is delivered over two events. The final time of $T = 3$ days was used within simulations so that computation was not prohibitively time consuming, and the two different total rainfall amounts were arbitrarily set to realistic average 3-day values with a sufficient difference between them to make the prospect of contrasting simulation results more likely. In each simulation, initial water content was set to a uniform value of $\theta(h_0) = 0.069$, which was low enough to ensure that the uptake of the root systems would increase as precipitated water entered the rooted soil, but sufficiently far from residual water content so that the wetting fronts

that formed during infiltration did not threaten the stability of the numerical scheme. Additionally, since the interest of these simulations was the longer term hydraulic effects of rhizodeposits, as opposed to the impact of their sudden addition to the soil, it was deemed acceptable to assume that, in the simulations that involved rhizodeposits, the exchange process between c_w and c_d was initially at equilibrium. According to the formula for rhizodeposit mass conservation (Schnepf et al., 2012) it was therefore set that:

$$\begin{aligned}\theta(h)c_{w,0} + \rho c_{d,0} &= 1 \quad x \in (-L, 0), \\ \partial_t c_d &= 0 \quad x \in (-L, 0), \quad t = 0, \quad (23)\end{aligned}$$

where the round value of 1 mg cm^{-3} was chosen to ensure that, according to formulae (20)–(22), the resulting values of $c_{w,0}$ and $c_{d,0}$ would have a noticeable effect on the surface tension and viscosity of the soil water solution and its contact angle with the soil pore surface. As a result, in the region of the soil occupied by roots, the initial concentrations of rhizodeposits in solution and rhizodeposits adsorbed to the pore surface were $c_{w,0} = 2.2 \text{ mg cm}^{-3}$ and $c_{d,0} = 5.1 \times 10^{-4} \text{ mg mg}^{-3}$, respectively. An initial condition of no rhizodeposit presence was set in the part of the domain without roots: $c_{w,0} = 0 \text{ mg cm}^{-3}$ and $c_{d,0} = 0 \text{ mg mg}^{-1}$. The 36 simulations were all carried out and results were recorded to ascertain the effect of rhizodeposit presence on soil water fluxes, evaporation at the upper soil surface, deep percolation at the lower boundary of the rooted zone, and root water uptake.

2.4 | Computations

Numerical solutions for pressure head, rhizodeposits in solution, and rhizodeposits adsorbed to the pore surfaces were obtained from the model (1)–(4) using a conformal finite element method and an implicit Euler discretization in time. The intervals of the one-dimensional domain discretization were each 0.593 cm in length, and the time step size was 0.001 day. Linearization of the functions for water content and hydraulic conductivity in (1) was carried out using the L-scheme, see for example, List and Radu (2016). The numerical method was implemented using the FEniCS library (Alnæs et al., 2015) and the algorithms to construct NRLD and SAD from CRoot-Box architectures were the same as those used by Mair et al. (2022, 2023) and run in Python 3 using the libraries NumPy and SciPy (Harris et al., 2020). The visualizations of one-dimensional root density profiles, water distributions, and flux strength were generated in Paraview (Ahrens et al., 2005). The plots in Figure 2, showing the architectures of the studied root systems, were generated using code of Dupuy et al. (2005), which was run in MATLAB 2024b. The Bayesian

optimisation algorithm applied by Gómez et al. (2025) to calibrate model (2.1)–(2.4) was implemented using the scikit-optimize library in Python 3 (Head et al., 2018). The codes that were used to run all algorithms, along with documentation for their implementation, can be found in the repository associated to the paper (Mair, 2025). This repository also contains all the data that were generated from the simulations.

3 | RESULTS

3.1 | Effect of rhizodeposits on water infiltration

For the effect of rhizodeposits on water infiltration, and the subsequent consequences regarding root water uptake, we focus only on simulations from two of the six precipitation regimes that were considered. Namely the six scenarios with events3total12 and the six with events1total28. These scenarios sit at opposite ends of the spectrum regarding the observed effect of rhizodeposits on root water uptake, with the presence of rhizodeposits most positively affecting total root water uptake in scenarios with events3total12, where the amount of water delivered per precipitation event was smallest, but negatively affecting uptake in some scenarios with events1total28, where the amount of water per precipitation event was largest. Within the other simulation scenarios it also generally held that as the amount of water delivered per precipitation event increased, the influence of rhizodeposits on total root water uptake became less positive, but the observed effect was more modest. Therefore, in the interest of brevity, the decision was made to limit the results displayed in Figures 3 and 4 to these two precipitation regimes with events3total12 and events1total28, where the contrast in the effect of rhizodeposits on water infiltration and root water uptake was most stark.

In all six of the scenarios with events3total12, simulations showed that downward water flux was greater in all scenarios where rhizodeposits were present compared to when they were not (Figure 3). Water infiltration increased with rainfall both with and without the presence of rhizodeposits. However, the increased infiltration in the presence of rhizodeposits became more and more pronounced with each rainfall event (Figure 3). At all observed time points, and for each age of root system, this increased water infiltration in the presence of rhizodeposits lead to a higher water content at the soil depth where root length density was highest (Figure 3). Moreover, at no point of the simulation did this enhanced water infiltration appear to have caused much movement of water into deeper non-rooted soil layers.

In the six scenarios with events1total28 (Figure 4), downward water flux was greater than in those with events3total12 (Figure 3). Like with events3total12, however, in the sim-

ulations with events1total28 the downward flux of water was always greater with rhizodeposits present than without (Figure 4). Similarly true in all these cases was that at 0.5 days, just after the single/first rainfall event, the increased infiltration of water had resulted in an increased water content in the depths of soil where the root length density of all root systems was highest. In the cases with events3total12, the magnitude of the downward water flux increased over the simulation (Figure 3). In contrast, though, with events1total28, the magnitude of the downward flux increased to its peak at 0.5 days and then decreased over the rest of the simulation (Figure 4). Furthermore, for the cases with age6events1total28rhizY and age15events1total28rhizY, the initial increase in infiltration was excessive and after 1.5 days had already resulted in a considerable amount of water being lost to depths with low or zero root length density (Figure 4). Since the roots of the 30-day old root system were allocated more equally and to greater depths of the soil, the increased infiltration in the scenario age30events1total28rhizY compared to age30events1total28rhizN did not reduce the availability of water for uptake in the same way that it did for the younger plants.

3.2 | Effect of rhizodeposits on rates of evaporation, deep percolation, and root water uptake

For all root systems in all precipitation conditions, and at all simulation time points, evaporation rates were lower with rhizodeposits present in the soil than without (Figures 5 and 6d–f). It was also the case for all precipitation regimes that, following a rainfall event, the water uptake rates of all root systems in soils with rhizodeposits became larger (for at least a finite time period) than those in soils without rhizodeposits (Figures 5 and 6j–l). For the 30-day old root system, which had the deepest rooting depth, this higher uptake rate in the presence of rhizodeposits was observed at all points in time, and this was true across all precipitation regimes (Figures 5 and 6j–l). Moreover, at all time points of the scenarios with age6total12 or age15total12, where plants had shallower root systems, the uptake rates were also higher with rhizodeposits present than without (Figure 5j–l). However, it can be seen from the scenarios age6events1total12rhizY and age6events1total12rhizN that with rhizodeposits present, the rate of deep percolation from the rooted zone started to increase during the second half of the simulation (Figure 5g). Consequently, the difference under this precipitation pattern between the increased uptake rate of the 6-day old plant with rhizodeposits and the lower rate without rhizodeposits had almost disappeared by the end of the third day (Figure 5j). This, therefore, suggested that a higher uptake rate with rhizodeposits than without

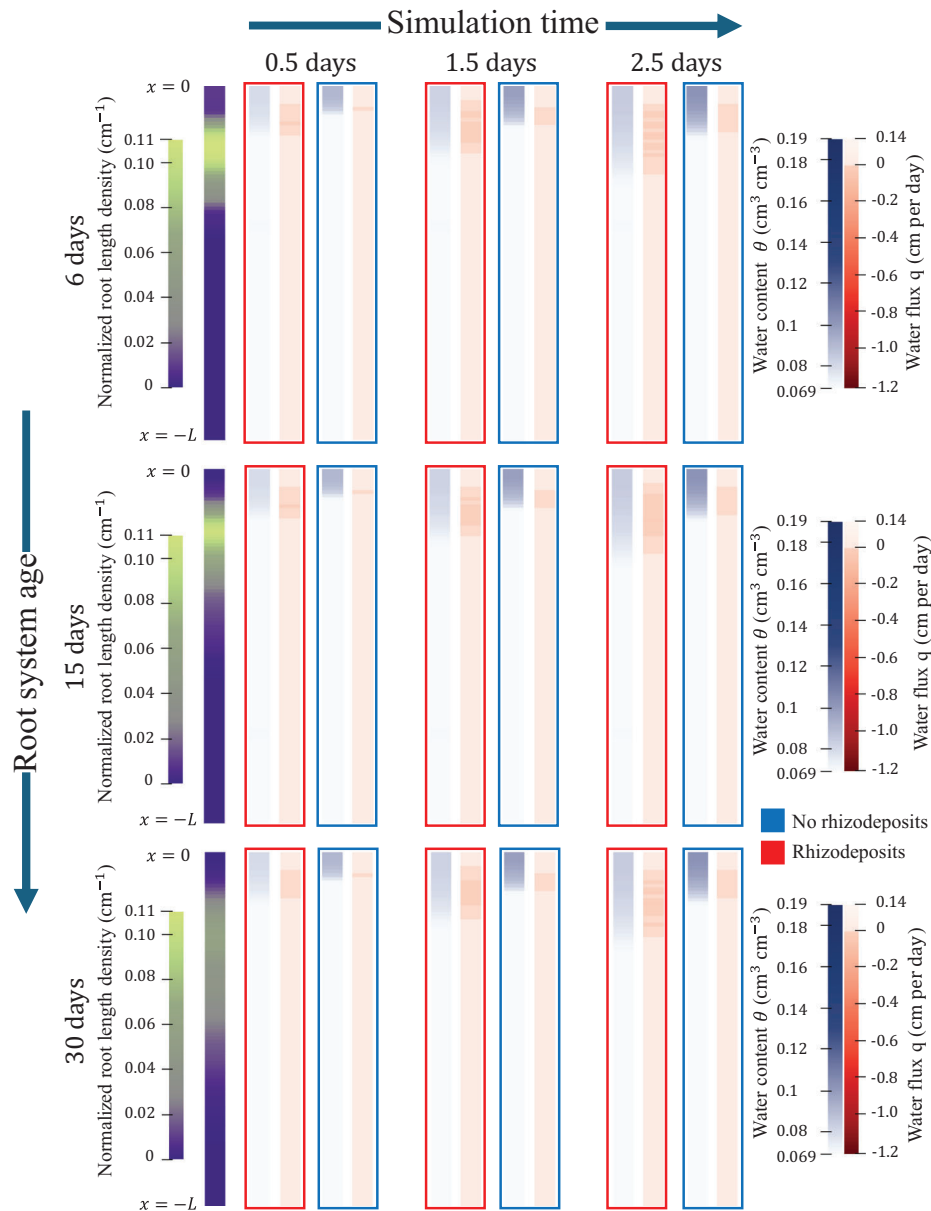


FIGURE 3 The influence of rhizodeposits on the evolution of soil water content and flux over time for the six simulated scenarios with age6events3total12, age15events3total12, and age30events3total12 (named according to the abbreviation system explained in Table 2). The figures of the leftmost column show the normalized root length density profiles of each plant within the 1-D soil domain (age increasing downward). The remaining columns show snapshots of the water dynamics within the domain at different time points of each simulation. Each row relates to soil vegetated by the roots of each age of plant. Negative values of water flux q indicate downward water flow through the domain and positive values indicate upward flow.

may not necessarily be maintained for the entire simulation under all combinations of total precipitation, delivery pattern of precipitation, and root system age. Indeed, under all scenarios with 0.28 cm of total precipitation the shallower 6- and 15-day old root systems initially assumed higher uptake rates with rhizodeposits in the soil but, eventually, the uptake rates in soil without rhizodeposits overtook and maintained higher values until the end of the simulation (Figure 6j–l). The timing of this overtake can be seen to approximately

coincide with the times at which the rates of deep percolation from the rooted zones containing rhizodeposits increased away from zero (Figure 6g–i), and was earliest for the precipitation pattern in which all 0.28 cm of water is delivered in one event.

Finally, within certain time intervals of simulations, the uptake rates of the 15-day old root system were slightly higher than those of the 30-day old, and this was true regardless of whether rhizodeposits were present or not (Figures 5 and 6).

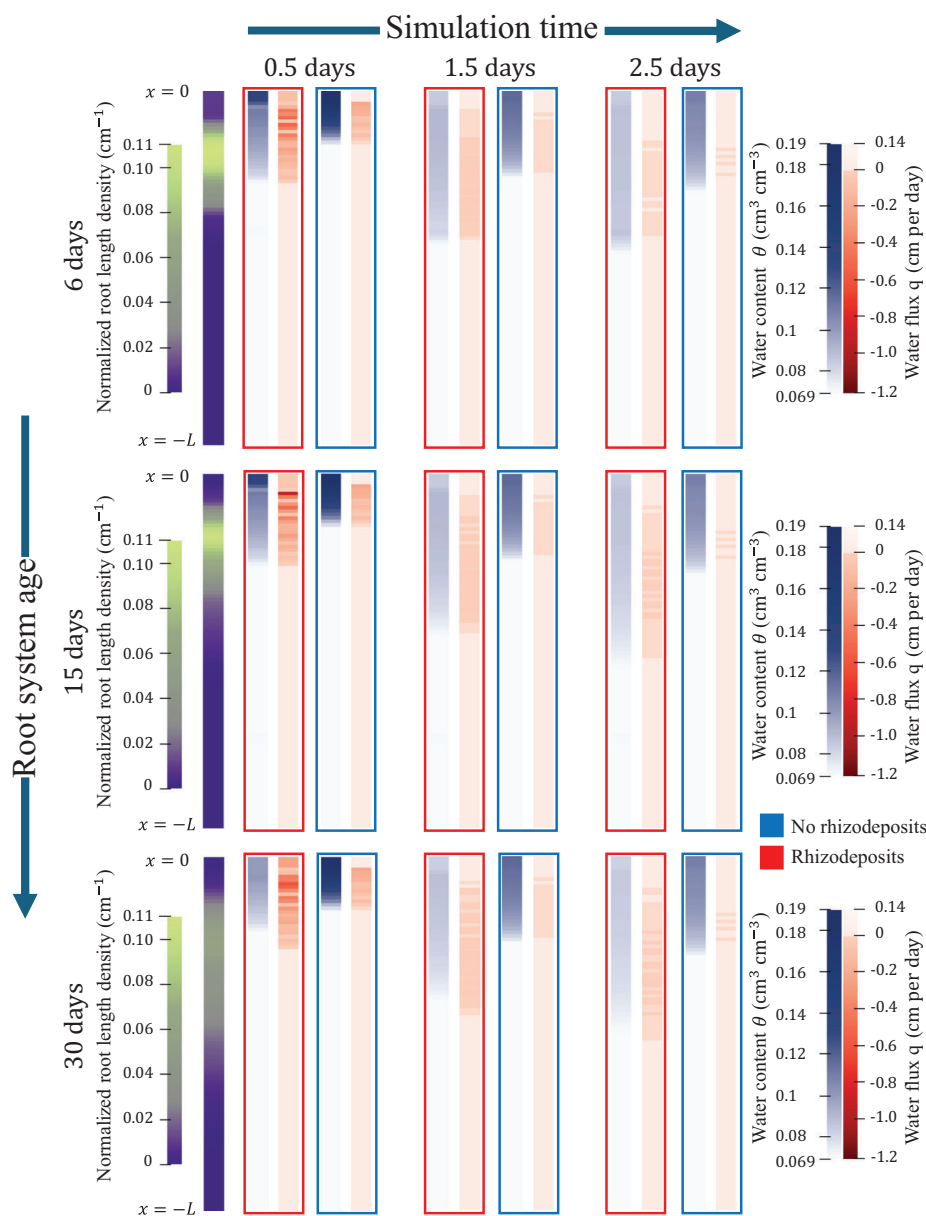


FIGURE 4 The influence of rhizodeposits on the evolution of soil water content and flux over time for the six simulated scenarios with age6events1total28, age15events1total28, and age30events1total28 (named according to the abbreviation system explained in Table 2). The figures of the leftmost column show the normalized root length density profiles of each plant within the 1D soil domain (age increasing downward). The remaining columns show snapshots of the water dynamics within the domain at different time points of each simulation. Each row relates to soil vegetated by the roots of each age of plant. Negative values of water flux q indicate downward water flow through the domain and positive values indicate upward flow.

3.3 | Rhizodeposit effect on total uptake and the sensitivity to precipitation pattern and root system maturity

In all rainfall patterns with 0.12 cm total precipitation, total water uptake of every age of root system was greater if rhizodeposits were present in the soil than if they were not (Table 3). Moreover, for all rainfall patterns of 0.28 cm

total precipitation, the total uptake of the 30-day old root system was also always larger when rhizodeposits were present than when they were not (Table 3). These results are expected given that, as previously mentioned, in these cases the higher uptake rates in the presence of rhizodeposits were maintained over the entire simulation period. The more nuanced results come from the cases where uptake rates with rhizodeposits in the soil were at some stage overtaken

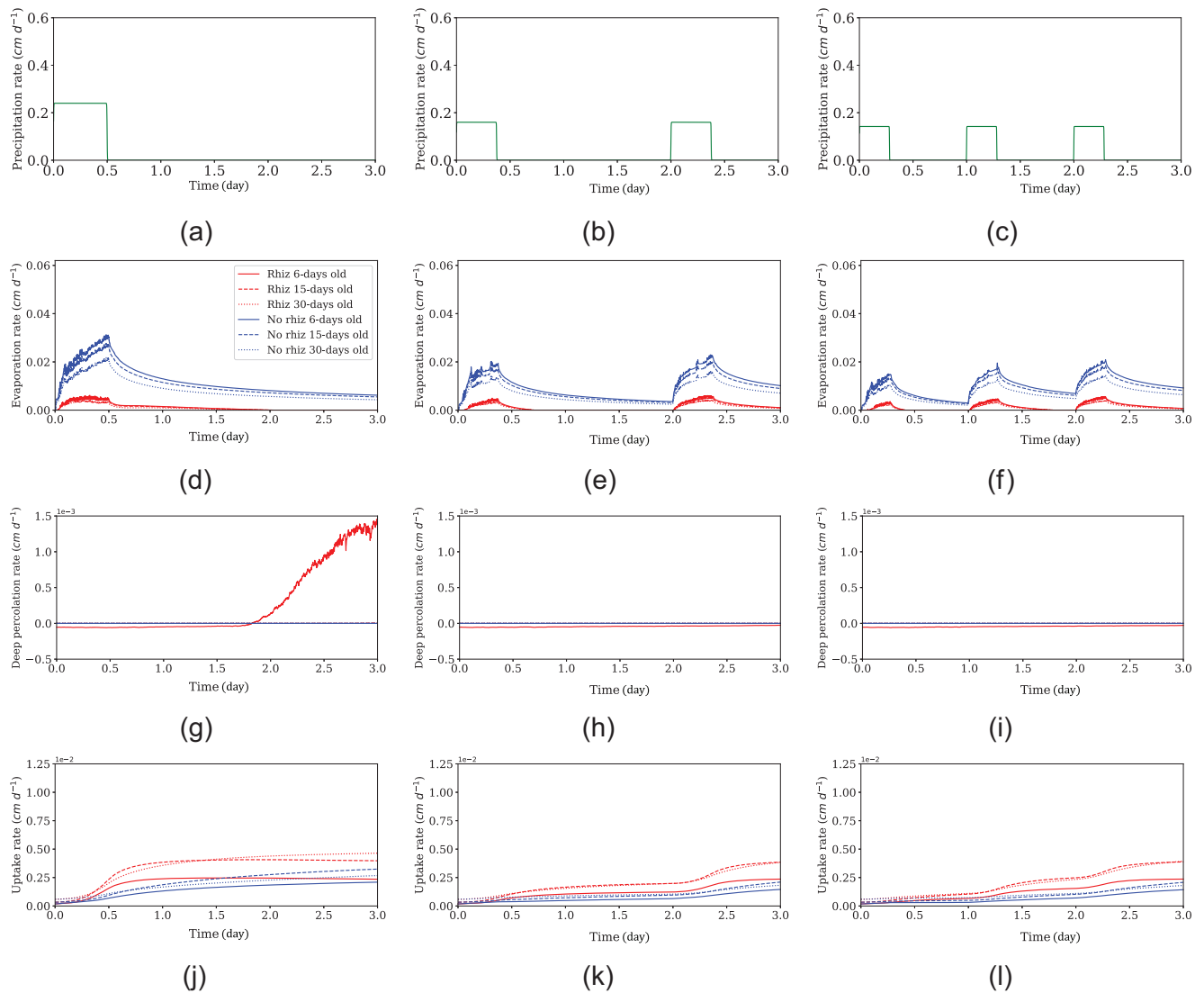


FIGURE 5 The influence of rhizodeposits on evaporation, deep percolation, and root water uptake under the different delivery patterns of lower total precipitation (0.12 cm). (a–c) Precipitation patterns considered. (d–f) Evaporation rates for each pattern and for soils occupied by each age of root system with or without rhizodeposits present. (g–i) Corresponding deep percolation rates. (j–l) Corresponding uptake rates.

by the uptake rates without rhizodeposits. Specifically, in the scenarios with `age6events2total28`, `age15events2total28`, `age6events3total28`, and `age15events3total28`, the total root water uptake in the simulations with rhizodeposits was higher than in the corresponding simulations without rhizodeposits (Table 3). However, in the scenarios with `age6events1total28` and `age15events1total28`, the uptake rates in the scenarios with rhizodeposits were overtaken at an earlier time point by the rates in the corresponding scenarios without rhizodeposits (Figure 6j), and the total uptake with rhizodeposits in the soil ended up being lower than if the rhizodeposits had not been present (Table 3).

This result can also be seen in Figure 7, which shows that the percentage change in total water uptake resulting from the presence of rhizodeposits was negative in the cases

of `age6events1total28` and `age15events1total28`. Considering root systems of all ages, there was, in fact, a general trend that the percentage positive change in total water uptake from including rhizodeposits in the simulations decreased as the amount of total rainfall in the 3-day period was increased (Figure 7). In addition, when fixing total rainfall and considering the pattern by which it was delivered, it is also true that the positive effect of rhizodeposits on total water uptake was reduced as the precipitation was delivered over fewer events (Figure 7).

Regarding the increased uptake rates of the 15-day old root system compared to the 30-day old root system, which were observed during certain intervals of specific precipitation scenarios, the net effect on total uptake can also be seen in Table 3. In the end, there were six

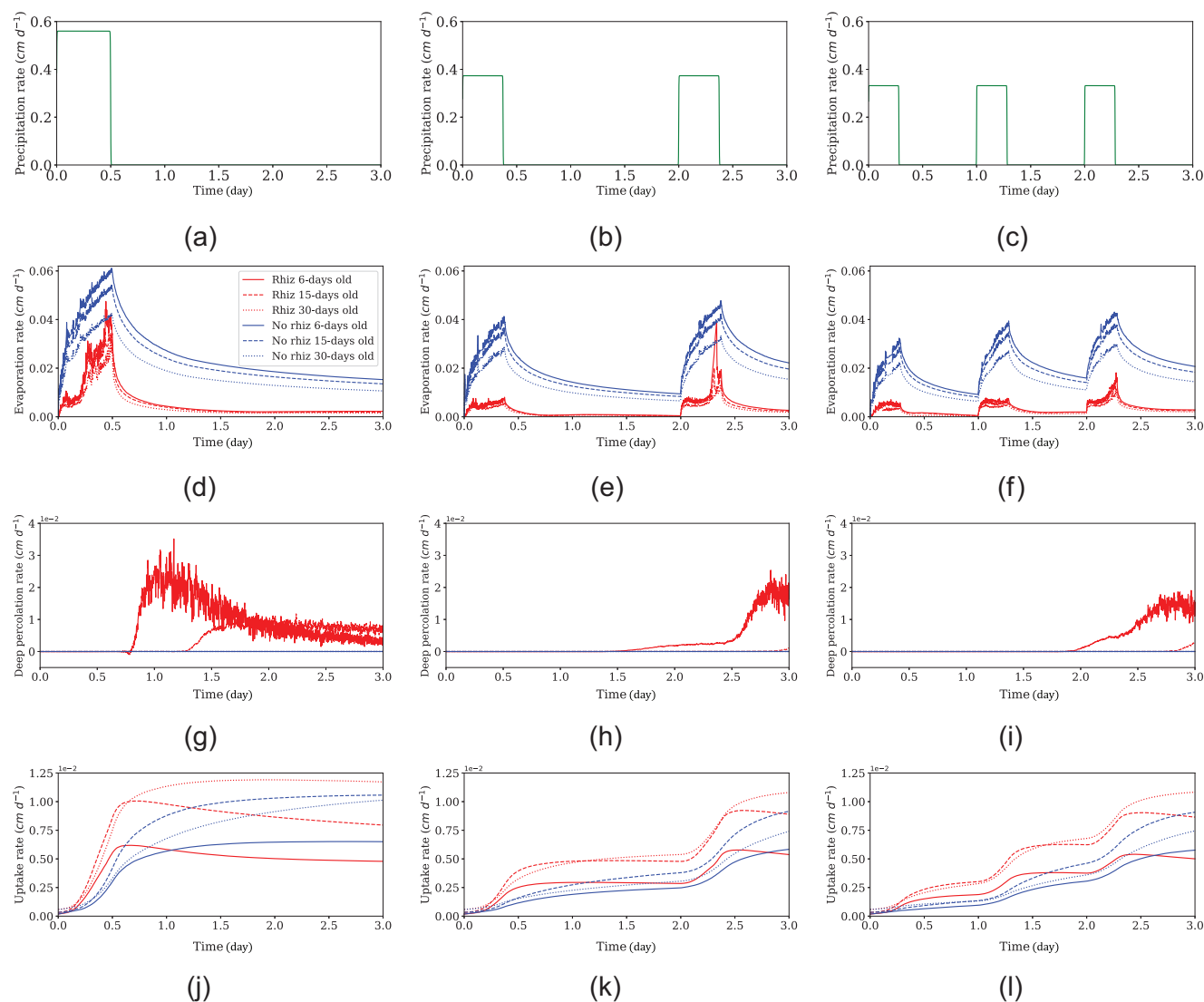


FIGURE 6 The influence of rhizodeposits on evaporation, deep percolation, and root water uptake under the different delivery patterns of higher total precipitation (0.28 cm). (a–c) Precipitation patterns considered. (d–f) Evaporation rates for each pattern and for soils occupied by each age of root system with or without rhizodeposits present. (g–i) Corresponding deep percolation rates. (j–l) Corresponding uptake rates.

instances where the 15-day old root systems total uptake was greater than or equal to the 30-day old root system. Namely, when total precipitation was 0.12 cm and all delivered in a single event with no rhizodeposits in the soil (events1total12rhizN), when total precipitation was 0.12 cm and delivered over two or three events with rhizodeposits in the soil (events2total12rhizY and events3total12rhizY), and for all scenarios with 0.28 cm total precipitation and no rhizodeposits in the soil (events1total28rhizN, events2total28rhizN, and events3total28rhizN). These results come from an effect of using a macroscopic uptake function based on normalized root length density to compare performance of root systems of different ages. To our knowledge, this effect has not been previously reported and, as such, will be explained further in Section 4.2.

4 | DISCUSSION

4.1 | Rhizodeposits as a mechanism for facilitated water infiltration

As previously mentioned, rhizodeposits from roots of certain species have been shown to have a surfactant effect when present in the soil water solution (Naveed et al., 2019; Read & Gregory, 1997; Read et al., 2003). Since the application of surfactants in agriculture often aids water infiltration into hydrophobic soils (Ogunmokin et al., 2020), it is plausible that facilitated water infiltration observed in vegetated soil (Cerdeira, 1999; Marshall et al., 2014) could be partly attributed to the presence of rhizodeposits that have a predominantly surfactant-like effect.

TABLE 3 Totals for evaporation at the soil surface, deep percolation at the lower boundary of the rooted zone, and root water uptake for each root system, under each precipitation regime, with and without rhizodeposits. Rows and columns are labelled for each simulation according to the abbreviation system explained in Table 2. Negative values in the deep percolation column indicate that there was a net inflow of water through the lower boundary of the rooted zone as a result of capillary rise.

	Evaporation (cm)		Deep percolation (cm)		Root water uptake (cm)	
	rhizN	rhizY	rhizN	rhizY	rhizN	rhizY
events3total12						
age6events3total12	30.54×10^{-3}	4.37×10^{-3}	0.003×10^{-5}	-0.13×10^{-3}	1.91×10^{-3}	3.76×10^{-3}
age15events3total12	27.05×10^{-3}	3.94×10^{-3}	0.003×10^{-5}	0.001×10^{-3}	2.78×10^{-3}	5.98×10^{-3}
age30events3total12	21.24×10^{-3}	3.11×10^{-3}	0.003×10^{-5}	0.001×10^{-3}	3.08×10^{-3}	5.88×10^{-3}
events2total12						
age6events2total12	30.19×10^{-3}	4.21×10^{-3}	0.003×10^{-5}	-0.13×10^{-3}	1.99×10^{-3}	3.73×10^{-3}
age15events2total12	26.75×10^{-3}	3.74×10^{-3}	0.003×10^{-5}	0.001×10^{-3}	2.89×10^{-3}	5.94×10^{-3}
age30events2total12	21.00×10^{-3}	2.96×10^{-3}	0.003×10^{-5}	0.001×10^{-3}	3.16×10^{-3}	5.83×10^{-3}
events1total12						
age6events1total12	36.60×10^{-3}	3.77×10^{-3}	0.003×10^{-5}	0.79×10^{-3}	4.31×10^{-3}	6.24×10^{-3}
age15events1total12	32.42×10^{-3}	3.33×10^{-3}	0.003×10^{-5}	0.002×10^{-3}	6.39×10^{-3}	10.19×10^{-3}
age30events1total12	25.46×10^{-3}	2.42×10^{-3}	0.003×10^{-5}	0.001×10^{-3}	5.59×10^{-3}	10.59×10^{-3}
events3total28						
age6events3total28	67.32×10^{-3}	10.67×10^{-3}	0.003×10^{-5}	8.76×10^{-3}	7.51×10^{-3}	9.56×10^{-3}
age15events3total28	59.63×10^{-3}	9.41×10^{-3}	0.003×10^{-5}	0.24×10^{-3}	11.26×10^{-3}	15.72×10^{-3}
age30events3total28	46.83×10^{-3}	7.04×10^{-3}	0.003×10^{-5}	0.001×10^{-3}	9.27×10^{-3}	16.68×10^{-3}
events2total28						
age6events2total28	65.34×10^{-3}	10.40×10^{-3}	0.003×10^{-5}	8.48×10^{-3}	7.69×10^{-3}	9.77×10^{-3}
age15events2total28	57.88×10^{-3}	8.42×10^{-3}	0.003×10^{-5}	0.08×10^{-3}	11.54×10^{-3}	15.86×10^{-3}
age30events2total28	45.45×10^{-3}	6.55×10^{-3}	0.003×10^{-5}	0.001×10^{-3}	9.40×10^{-3}	16.69×10^{-3}
events1total28						
age6events1total28	78.60×10^{-3}	18.64×10^{-3}	0.003×10^{-5}	23.33×10^{-3}	15.63×10^{-3}	14.30×10^{-3}
age15events1total28	69.63×10^{-3}	16.21×10^{-3}	0.003×10^{-5}	12.73×10^{-3}	24.55×10^{-3}	23.96×10^{-3}
age30events1total28	54.68×10^{-3}	12.62×10^{-3}	0.003×10^{-5}	0.001×10^{-3}	24.55×10^{-3}	30.12×10^{-3}

In our simulations where rhizodeposits were present, water infiltrated more quickly after the onset of precipitation and water loss by evaporation was reduced (Figures 5 and 6; Table 3). This concurs with similar experimental results regarding evaporation losses under the application of bio-surfactants (Gutierrez et al., 2022). However, our results also highlighted that, in terms of water availability to the root system under enhanced infiltration, a clear trade-off exists between reduced evaporation losses and increased deep percolation losses (Table 3).

The net effect of this trade-off will vary with soil type, precipitation pattern, and root system morphology. In very conductive sandy soils, where water retention is comparatively poor (Bhardwaj et al., 2007), low irrigation rates are recommended for optimizing the fraction of total irrigation that can be taken up by roots (Alhammadi & Al-Shrouf, 2013), since increases to infiltration rate only serve to reduce the time that the water is available for uptake. In contrast, for less conductive soils, water losses from evaporation are

likely to be larger than from deep percolation and, therefore, increases to infiltration rate improve root water uptake and crop yield (Amooh & Bonsu, 2015; Oostindie et al., 2010).

In terms of precipitation pattern, Chaichi et al. (2015) found that if rainfall or irrigation events were smaller and total water was delivered more gradually, then the effect of enhanced infiltration on root water uptake and yield was positive. This is also reflected in our simulations where with a lower total precipitation delivered over three small rainfall events, the increase to downward water flux caused by rhizodeposits allowed more water to reach the rooted zone and be taken up by the plant (Figures 3 and 5; Table 3). However, if total rainfall is concentrated into fewer but more intense events, then the enhancement of downward flow can become less beneficial or even detrimental to the availability of water to the root system (Blackwell, 2000).

Our results (Figure 7) showed, for root systems of all ages, that rhizodeposit-induced increases in infiltration were generally less beneficial to total root water uptake as the quantity

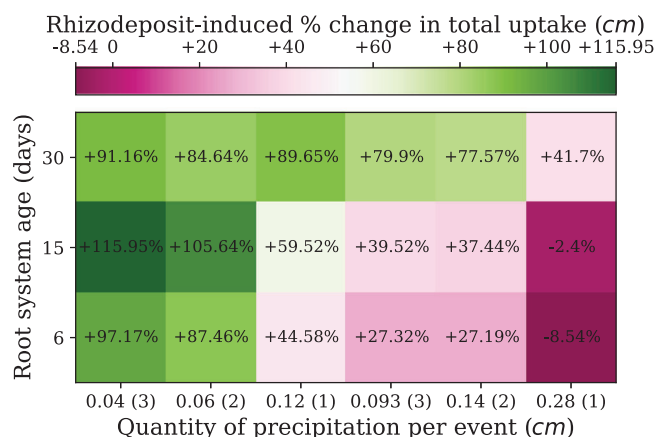


FIGURE 7 The strength of the positive or negative effect of rhizodeposits on total water uptake of a root system in relation to root system age and the quantity of precipitation delivered in each rainfall event of the precipitation regime. On the x-axis, numbers outside of brackets indicate the quantity of rainfall in one event of the regime. Numbers in brackets show the total number of events in that regime.

of rainfall per event increased. Moreover, when all 0.28 cm of precipitation was delivered in one rainfall event, the total uptake of the 6- and 15-day old root systems was actually worsened by the presence of rhizodeposits (Figure 7). That is, despite the presence of rhizodeposits initially accelerating the water's arrival to the root zone and improving uptake rates (Figure 6), ultimately the larger downward flux of water meant that the window of time it was available for uptake by these shallower root systems was shorter than when rhizodeposits were not present (Figure 4). In contrast, however, the deeper roots of the 30-day old root system meant that within the simulations the rhizodeposit-induced increases to infiltration never negatively affected uptake, regardless of precipitation pattern (Figure 7). Such a result concurs with the experimental findings of Shao et al. (2009) where, due to their larger root systems, the water uptake of mature wheat root systems appeared robust under alterations to irrigation pattern.

A number of mechanisms have been attributed to how reductions to surface tension and the presence of surfactants can enhance soil hydraulic conductivity and, thus, alter the amount of water that infiltrates into rooted soil post-precipitation and the length of time it is retained there (Henry & Smith, 2003). First, under reduced surface tension, the water content at a given pressure head during wetting is increased (Henry & Smith, 2003), thus promoting an increase in the hydraulic conductivity at the same pressure head. Furthermore, reductions to surface tension promote the reconnection of thin water films between pores (Tokunaga, 2009) and surfactants can also aid in the declogging of pores (Tumeo, 1997). Both these processes improve hydraulic

connectivity within the soil and, in turn, increase hydraulic conductivity (van Genuchten & Pachepsky, 2011).

Such an effect is captured within the formulation of our water transport model (1), where rhizodeposit-induced reductions to surface tension decrease the pressure head for a given water content when wetting and increase the soil hydraulic conductivity. It results in a trade-off between increasing root water uptake through improved infiltration of water into the rooted zone and decreasing uptake from the acceleration of downward drainage away from roots.

The incorporation of rhizodeposit-induced reductions to soil water surface tension in model (1) also explains why in the simulations of age6events3total12rhizY and age6events2total12rhizY, there was a net capillary rise into the rooted zone when rhizodeposits were present (Table 3). In these cases, the lower boundary of the rooted zone corresponded to the 6-day old root system and, hence, was closer to the upper soil surface where smaller quantities of water were being delivered per precipitation event. This meant that infiltration to the region around the lower boundary of the rooted zone was gradual, and the water contents at either side of this boundary were both increased and remained similar throughout the simulation. However, during time periods when the soil on both sides of the lower rooted zone boundary was wetting, the effect of rhizodeposits in reducing the soil water surface tension would have resulted in a lower pressure head on the rooted side (Figure 1b). Therefore, a gradient in potential was created from below to above the lower boundary of the rooted zone and, hence, caused a capillary rise of water in the upward direction and into the rooted zone. This did not happen for larger root systems because in environments with lower total precipitation, water did not reach the lower boundaries of the rooted zone and, hence, water contents remained low and any shifts in the retention curve caused by rhizodeposits had a much smaller effect.

4.2 | Advancing the incorporation of plant root influence into models of soil hydraulics

As models of soil water transport and root water uptake are developed further in the future, several factors will need to be kept in mind so that the accuracy with which they incorporate the effect of plant roots on soil hydraulic properties continues to improve. One such factor is that rhizodeposits from different species do not have the same impact on soil hydraulics. For instance, the effect on viscosity of maize rhizodeposits that was reported by Naveed et al. (2019) was stronger than the effect that was observed by Hosseini and Mosaddeghi (2024) for wheat rhizodeposits. As a result, models like (1) may predict different outcomes for their effect on the availability of water for root uptake under varying precipitation regimes. Furthermore, the characteristics of rhizodeposits are not

constant over time but change according to plant growth stage and activity of the rhizosphere microbial community (Tian et al., 2020). Therefore, if the degradation of rhizodeposits by microbes was incorporated into models like (1)–(4) and that of Landl et al. (2021), then it is possible that simulations would give significantly different results regarding the effect of rhizodeposit presence on soil hydraulics and root water uptake rate.

Ensuring that root water uptake and evapotranspiration are modeled in an accurate way is also important. In model (1), no variance of the reference evapotranspiration ET_0 with changing atmospheric humidity was accounted for. Instead, the boundary condition for evaporation only varied according to soil water content at the surface and, therefore, did not account for any reduction to evapotranspiration that would likely occur during rainfall due to increased atmospheric humidity. Because of this, simulated evaporation rates were highest when surface water content was highest, which was during precipitation events. However, coupling model (1) to a model for atmospheric water dynamics would have increased model complexity considerably, bringing a large number of new parameters values to identify and making efficient model simulation computationally challenging. If atmospheric humidity was incorporated, then model simulations would likely have shown a greater level of infiltration into the soil during rainfall and, hence, lower values of total evaporation losses but possibly higher levels of deep percolation losses. Nevertheless, the trade-off between reduced evaporation losses and increased deep percolation losses would likely still have been present. Moreover, since manually supplying water to the soil is not associated with increased atmospheric humidity like precipitation is, the results from model (1) in its current form also remain informative if it is assumed that water is being added to the soil via irrigation as opposed to crops being rain-fed.

In terms of root water uptake, the sink term employed in (15) is an empirical macroscopic root water uptake function like that described by Simunek and Hopmans (2009). These functions combine terms for root length density, local water stress, and transpiration, and their benefit is that they are sensitive to local soil water content and incorporate compensatory uptake, while avoiding the computational demands of explicitly modeling flow into and through the entire root system. There are, nevertheless, known disadvantages to macroscopic uptake functions. First, their parameters represent high level concepts and, hence, are difficult to measure experimentally (Cai et al., 2018). Second, if two regions have equal root length density, and their water contents are above the given critical level that indicates water stress, then the uptake from these two regions does not depend upon water content (Vanderborght, Couvreur et al., 2024). Moreover, traditional macroscopic functions cannot capture hydraulic lift (Javaux et al., 2013). The results of this work, however, have shown another potential flaw in macroscopic uptake

models such as (15). Namely, in six of the simulated scenarios, the 15-day old root systems total water uptake was greater than or equal to the uptake of the 30-day old root system. The issue arises from the normalized root length density function NRLD in (15), which is used to determine a soil region's maximum possible contribution to total uptake and computed by dividing the root length density function by the system's total root length. This formulation means that as a root system grows larger, and the roots within a given region make up a smaller proportion of total root length, the value of NRLD for that region decreases (see Figures 3 and 4). As a result, even if the water content of a given soil layer and the conductance of the root system remain constant, the simulated uptake from the roots in that layer may well decrease as the root system grows. The consequence of this is that a younger root system may exhibit higher rates and cumulative totals of simulated uptake than a mature root system, despite the younger plant having a much lower total root length within the wet regions of soil (Figures 5 and 6; Table 3). As a potential answer to this problem and the others mentioned, Vanderborght, Leitner et al. (2024) have recently proposed an alternative formulation of macroscopic uptake function where the root system is represented in a network format. However, unlike several previous studies that also take this approach (Couvreur et al., 2012; Doussan et al., 1998; Javaux et al., 2008; Somma et al., 1998), the method of Vanderborght, Leitner et al. (2024) does not solve a resulting linear system but, instead, uses the network to determine the contribution of each root to total root system conductance. By accounting for transpiration demand, root collar water potential, and local soil water potential, these conductance values then determine the uptake from each soil layer. Through the application of homogenization theory to a system of coupled equations for root and soil water transport, Mair and Ptashnyk (2024) also recently derived a macroscopic equation for soil water transport where the term for root water uptake incorporates microscopic information like local hydraulic conductance and xylem water potential. In future, this could also be a viable method of formulating a macroscopic uptake function that is free from the issues raised previously.

Effective development of future models will rely upon advancing the methods by which data can be collected to determine parameter values. In real soil it would be very challenging to determine the parameters in model (1)–(4) that govern the extent to which rhizodeposits increase hydraulic conductivity, and the rates at which they adsorb and desorb to and from soil. However, the transparent properties of the Nafion particles used in place of real soil in the experiments of Gómez et al. (2025) allow the collection of highly resolved data during water infiltration and, hence, made it possible to approximate these values. Nafion particles do not fully replicate the heterogeneity of a real soil structure and it is true that this inevitably limits the inferences that can be made

from models that are parametrized using data from synthetic soil microcosms like these. Nevertheless, the simulations that were carried out using model (1)–(4) have provided an indication that the effect of infiltration-facilitating rhizodeposits on the availability of water to root systems may be sensitive to precipitation pattern and root system maturity. This, at the very least, has identified a potentially fruitful area of further investigation and provided motivation for the design of an experimental system that will allow the parametrization of models like (1)–(4) for real soils.

4.3 | Harnessing rhizodeposits for the development of crop ideotypes with resilience to environment-specific stresses

The Green Revolution of the 1960s, with the increased use of chemical fertilizers, pesticides, and precise irrigation, shifted the attention of crop breeders toward the question of maximizing yield under conditions where stresses from water and nutrient shortages or competition were minimal (Preece & Peñuelas, 2020). This approach led to a focus on optimal aboveground traits, where one direction was to breed for crops that maximized the ratio of harvestable aboveground biomass to total aboveground biomass (the harvest index) (R. Richards et al., 1993). Since progress in this aspect rapidly became marginal, improvements to total aboveground biomass were then sought, and achieved, through breeding for rapid leaf area development to reduce evaporation losses from the soil and enhance leaf-level water use efficiency. Within the context of roots, however, the aim of many was to maintain root development at the necessary minimum so as to maximize resource allocation to harvestable biomass under ideal growing conditions (Condon et al., 2004). Nevertheless, with the increasing incidence of global drought, breeding strategies for resilient crops have latterly started to consider more developed root systems with features that improve access to soil water.

Genes have been identified that control root gravitropic response and radial growth (Hafeez et al., 2024), and there is a belief that increasing rooting depth should be the target of breeders. This is, first, because it allows rain-fed crops to access deep stores of water during periods of drought (Lynch, 2013; Uga et al., 2013) and, second, because in wheat systems root arrival at deep water stores tends to coincide with grain-filling season thus boosting preharvest biomass (Ober et al., 2021; Palta et al., 2011; Wasson et al., 2012). In addition to a root system's architecture, there is growing data to suggest that the production rate and chemical profile of rhizodeposits are adaptive to biotic factors like growth stage, and external factors such as soil type and water or nutrient availability (Rolfe et al., 2019; Wen et al., 2022; Williams & de Vries, 2020). This gives weight to claims that rhizodeposits can be deployed by a plant to engineer the soil in accor-

dance to their needs, as opposed to being a waste product whose potential effects on water and nutrient availability are merely coincidental. Since evidence of a genome effect on rhizodeposit and microbiome features has been shown (Iannucci et al., 2021), it has, therefore, been proposed that crops could be bred to promote the development of microbiomes that are specifically beneficial under certain stresses (Ober et al., 2021). For instance, as suggested by the findings of this work, if crop root systems are deep and soil water losses are predominantly from evaporation, then the production of rhizodeposits with surfactant like properties could improve root water uptake.

Despite this potential, however, few crop breeding programs currently incorporate rhizodeposit characteristics in their selection criteria (Preece & Peñuelas, 2020). One reason for this is that it is far from straightforward to verify that a given crop genotype produces rhizodeposits that affect soil characteristics in the way that they have been purported to. This is because nondestructively extracting rhizodeposits from roots grown in the field is massively challenging, and when taking the more simple approach of extracting from roots grown in hydroponic systems, it cannot be guaranteed that the same rhizodeposit properties would be observed for roots grown in soil (Oburger & Jones, 2018). Nevertheless, increasingly sophisticated methods are now being employed to overcome these hurdles (Casas & Matamoros, 2021) such as growing roots in soil then transferring to hydroponic systems for rhizodeposit extraction (Gargallo-Garriga et al., 2018; Herz et al., 2018), sampling rhizodeposits directly from soil by leachate collection (de la Paz et al., 2019), and the use of matrix-assisted laser ionization for more rapid mass-spectrometry analysis (Sasse et al., 2020). With these technological advances, knowledge of rhizodeposit properties continues to improve. Additionally, the emergence of synthetic soil microcosm systems (Mair et al., 2025) is facilitating the observation of the effects that these rhizodeposits have on soil hydraulic processes at the microscopic level (Hayat et al., 2021; Liu et al., 2025; Zarebanadkouki et al., 2019). Continued progress in this area will generate data that allow the improved formulation and calibration of models (like the one presented in this work) so that they better represent the impact of rhizodeposits on soil hydraulics in both a qualitative and quantitative sense. This will help to predict the effects of given rhizodeposits on metrics of crop performance and, hence, aid the selection of ideotypes for improved productivity and resilience to environmental stressors.

5 | CONCLUSION

In summary, a model for soil water transport was developed that incorporated the effect of wheat rhizodeposits on soil hydraulics. Simulations from this model were then carried out

to observe the effect of rhizodeposit-induced changes to soil water surface tension and viscosity, soil-water contact angle, and soil hydraulic conductivity on root water uptake. The findings of model simulations suggest that the consequences on root water uptake of rhizodeposits whose predominant effect is to enhance hydraulic conductivity and infiltration will depend upon the environmental conditions and RSA considered. For shallow root systems in environments where total precipitation is delivered in infrequent but heavy rainfall events, the enhancement of hydraulic conductivity may negatively affect uptake. On the other hand, results suggested that the total water uptake of deeper and more developed root systems would improve under such rhizodeposit-induced alterations to soil hydraulics. Indications were made as to how models like the one from this study may be further developed in the future, and how this may be facilitated by breakthroughs in technology for sampling the rhizosphere. Achieving these advances will not be trivial by any means, but it will help make the consideration of rhizodeposit properties within crop breeding programs a more feasible prospect.

AUTHOR CONTRIBUTIONS

Andrew Mair: Conceptualization; data curation; formal analysis; investigation; methodology; software; validation; visualization; writing—original draft; writing—review and editing. **Emma Gómez Peral:** Data curation; investigation; methodology. **Mariya Ptashnyk:** Methodology; supervision; writing—review and editing. **Lionel Dupuy:** Conceptualization; funding acquisition; investigation; methodology; project administration; resources; supervision; writing—review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data generated by all simulations, along with the codes to parametrize and numerically solve the model equations, are available from Zenodo via <https://doi.org/10.5281/zenodo.17106884> under a Creative Commons Attribution 4.0 International license.

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